

Predictive Decision Models for Multi-Channel Digital Marketing and Their Impact on Customer Relationship Management

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Abstract—This study aims to analyze the dimensions of digital marketing through multi-channel strategies and their role in strengthening customer relationships with the organization. This study derives its theoretical importance from the field of business administration, specifically marketing management using digital marketing, and the extent of the importance of multiple digital marketing channels in strengthening customer relationships with the institution. On a practical level, this study derives its importance from understanding the contribution of multiple digital marketing channels to strengthening customer relationships with Dubai NXN Post.

Keywords—decision models, digital marketing, customer relations

I. INTRODUCTION

Customer demands and innovations increasingly met in digital sales that require resilience of organization to target several channels. Multi-channel marketing normally used to implement exceptional competitive advantages to organizations [1]. With the development of the world of technology and information, which now dominates all sectors, a tremendous transformation has occurred in all aspects of cultural, social, and economic life. This is evidence of the opening of new horizons and future aspirations. It is clear that the methods of the modern technological revolution and the developments taking place in all fields have played a major role in bringing about radical changes in work methods and patterns in all fields [2]. Among these fields is marketing, which is one of the most technologically advanced sectors, and one of the most responsive to technological progress and innovation. It is also one of the most frequent users of modern and advanced technologies. These technologies have formed the axis of a marketing transformation that has resulted in a profound change in the way producers and consumers think [3].

From this point, marketing began to take on new dimensions to address these developments. It has become crucial in practice, serving as a key determinant of success for many global companies in various industrialized nations [4]. This is in addition to the challenges faced by organizations operating in local and global markets [5]. Digital marketing emerged as the modern marketing platform, its lower cost contributing to market expansion. Numerous companies specializing in digital marketing were established, striving to acquire the largest possible number of online customers interested in their services. It is now undeniable that all brands, companies, and sectors rely heavily on a diverse range of digital marketing channels to remain competitive in the fiercely competitive marketing landscape, particularly through the use of websites, social media platforms, YouTube, and mobile applications. Multi-channel strategies have variance form as shown in the figure 1 and it help the companies to presence in multi metods to be close to the customers [6].



Fig. 1. Multi channels marketing strategies.

Based on the above, we pose the following question: To what extent does the diversification of digital marketing channels contribute to strengthening customer relationships with Dubai NXNPost? Importance and Objectives of the Study: This study derives its theoretical importance from the field of business administration, specifically marketing management using digital marketing, and the extent of the importance of multiple digital marketing channels in strengthening customer relationships with the institution. On a practical level, this study derives its importance from understanding the contribution of multiple digital marketing channels to strengthening customer relationships with Dubai NXNPost.

II. METHODOLOGY OF THE STUDY

This study aims to address the role of multiple digital marketing channels in strengthening customer relationships with Dubai NXNPost. The first section covers the academic and conceptual framework of digital marketing, including a range of media and multiple channels. The second section examines and analyzes Dubai NXNPost's use of digital marketing channels, most notably: <https://www.nxn.ae/> - Dubia Post's website NXN-

BaridiMob - BaridiMob application- SMS (SMS)

III. LITERATURE REVIEW

The Concept of Digital Marketing: It is a marketing approach based on communicating with current and potential customers through digital content and internet channels, such as search engines, social media, and email. (Websites and online stores) (Hanan, 2022). It is also defined as promoting products derived from databases using distribution channels on the internet. (Gupta, 2012) in order to reach customers in a personal, effective, and timely manner. According to the American Marketing Association, digital marketing includes the activities and procedures simplified by P.K. Kannan [7] using digital technology to create, deliver, and provide value to customers and other partners.

Digital marketing also refers to creating and publishing content through digital media channels (websites, homepages, social media, email, and mobile applications) and promoting that content using a variety of strategies across paid digital channels, content sharing, pay-per-click (PPC), pay-per-click (SEM), and SEO. Owned, including social media, email, text, and other media [8].

The Concept of Digital Marketing Channels: In simple terms, marketing tools can be defined as a marketing tool, technology, strategy, or program used to promote a company's or brand's products or services and increase sales. Digital marketing tools can be used for online marketing and advertising. These tools rely on digital technology to reach the target audience and enhance the brand and increase sales [9]. Forms of Digital Marketing Channels: Digital marketing tools include a wide range of means and techniques, such as websites, e-commerce, search engine marketing, social media marketing, email marketing, content marketing, and marketing through Offers and

discounts, digital public relations marketing, and telemarketing, referred to as follows:

First: Website: Kotler [10] define a website as a collection of interconnected web pages forming a single entity managed by an individual or organization. This entity combines function, content, form, organization, and interaction, and is dedicated to one or more closely related topics.

Furthermore, customers judge a website based on criteria such as: ease of use, loading speed, clarity of the landing page, website appeal and page organization, appropriate use of colors and sounds, and respect for privacy and customer data confidentiality [11].

Several tools exist to optimize a website for search engine rankings, including those detailed in Table 1.

TABLE I. SEARCH ENGINE OPTIMIZATION TOOLS

Tools	Description
MOZ	One of the most popular digital marketing tools used by marketers, MOZ is considered a search engine optimization (SEO) tool that helps increase traffic. It is preferred by many, especially beginners, because it offers free services that include many tools used for search engine optimization without the need for a paid subscription. MOZ is a search engine optimization tool that helps increase traffic and is preferred by many.
Semrush	Semrush is one of the essential digital marketing tools that you can use to plan and monitor all your search engine optimization (SEO) strategies. What is the mechanism of digital marketing? (Al-Najadat, 2021) Google Keyword Planner: The competitiveness of these keywords.
Google Keyword Planner	Google Keyword Planner is one of the most important digital marketing tools that is indispensable for writing SEO-friendly content. It is used to determine the search volume for keywords that your audience is searching for, allowing you to target, evaluate, and provide other details.
Surfer SEO	Surfer SEO is a content writer-friendly tool that provides recommendations for the primary and secondary keywords that should be included in the content, the number of images, paragraphs, and lines in these paragraphs, as well as the prepositions, adverbs of time and place that should be included, and the relevant keywords. It also ensures that the content is free of any copying or duplication.

Second: Social Media: Social media has developed significantly in recent times and has become a priority marketing channel for all businesses. Social networks are an ideal place to gather data on what customers are searching for and preferring. They also enable organizations to exchange knowledge, benefit from customer opinions, and share ideas with each other (Neti, 2011). Managing all social media-related tasks manually is tedious and time-consuming, often without much real benefit. Many tools

exist that help accomplish the same tasks quickly and accurately. Some of the most important of these tools are detailed in table 2.

TABLE II. SOCIAL MEDIA MANAGEMENT TOOLS

Tools	Description
Hootsuite	Hootsuite is one of the most important digital marketing platforms, and a very important tool for managing social networks. Hootsuite offers numerous options for businesses ranging from small and medium-sized enterprises (SMEs) to large and leading companies. One of its key advantages is its centralized management system, making it ideal for businesses. It provides the largest selection of social media management tools for managing social media content from a single location, along with numerous tools to help individuals track and schedule all their accounts (Al-Najadat, 2021).
Buffer	Buffer is one of the most user-friendly social media management tools and is available at affordable prices. It is considered one of the most important social media management tools for small businesses. Through this tool, you can access many features, most importantly: (Mukhtar, 2022) - Scheduling content for all social media platforms. - Obtaining accurate engagement data and insights. - The ability to view all your posts from one place.
Crowdfire	Crowdfire is one of the best social media management platforms, offering a range of top features to help you use it. Boosting social media engagement and achieving tremendous growth will become much easier. Crowdfire In addition, the platform allows content publishing on blogs and enables interaction with customers to obtain excellent and fast customer service. It also allows discovering relevant content to share with an audience and even scheduling posts (to be published automatically at different times). (Mukhtar, 202)

Third: Mobile Marketing: Following the development witnessed in the telecommunications sector and the increased use of the internet via mobile devices, mobile marketing extends to text messages, which are considered an effective marketing tool by keeping customers informed of all new developments. Mobile marketing encompasses all customer-related activities conducted through the telephone, including the sale of products and services, the provision of information, and the confirmation of transactions, especially financial ones. It also includes a set of practices that allow organizations to communicate and engage

with their audience interactively via mobile phone, which is available on mobile messaging services. These messages consist of only 160 characters. (Antoine Lamarre) (MMS) Multimedia Messaging Services

Fourth: Mobile Applications: These are platforms that provide services to users in specific areas through standalone programs designed to run on mobile devices such as smartphones or tablets. They are considered a new marketing method (François Scheid, 2019) that allows for the creation of a new type of relationship. Mobile applications allow for the delivery of targeted and direct marketing content to users and can help build long-term relationships with customers by providing ongoing value. Some mobile applications require exclusive permissions from the user to access specific data on the smartphone, such as contact information or geographic location. This is the correct and required information for a specific user, which is not necessarily the same as the type of information that can be provided to a user located in another geographic area, especially in financial transaction services. Mobile applications are programs that require the user to download and install them on their smartphone. Applications are installed through the Apple App Store or Google Play Store.

Fifth: Content Marketing: This is a form of marketing that focuses on creating, publishing, and distributing content to the target audience online. Unlike other forms of online marketing, content marketing relies on anticipating and fulfilling the information needs of current customers. The core idea of content marketing is that a brand must give something of value to receive something of value in return, rather than Commercial advertising takes the form of an article instead of a banner ad when companies seek to market content.

The primary focus should be on the needs of potential customers. Once a customer's need is identified, information can be presented in a variety of formats, including news, videos, ebooks, infographics, email newsletters, case studies, Q&As, articles, and images. Digital Marketing Dimensions for Enhancing Customer Relationships: Digital marketing dimensions for enhancing customer relationships include a set of key aspects, the most important of which are: 1.3. Attraction: This involves attracting customers and engaging them through the organization's online presence, search engine rankings, and interactive advertising. Attraction includes: driving customers to the app, building an audience, and keeping the brand in mind [12].

Communication: This includes interacting and communicating with customers and providing added value to understand their reactions and satisfaction levels. Communication is also important for allocating time and marketing effort to multiple customers simultaneously.

Engagement: This involves placing customers in a state of interest and participation to achieve interaction with the services provided. This is done by providing valuable content that encourages users to complete actions (or business activities) [13].

Retention: One of the organization's goals is to retain customers and persuade them to repeat orders for the product or service. This is achieved by building positive relationships with

customers, updating the content of digital marketing tools, and providing fast information loading. Retention also requires advertising using digital marketing platforms and search engines.

Electronic marketing for promoting the elements of the digital marketing mix (product, distribution, promotion, pricing, operations, human element, physical facilities, productivity, and quality) [14].

Learning: This involves understanding customer details by gathering information about them and identifying their trends using surveys and questionnaires.

IV. APPLICATION OF DIGITAL MARKETING: CASE STUDY

The Role of Digital Marketing Channels in Enhancing Customer Relationships with Dubai NXNPost: Dubai NXNPost aims to provide comprehensive postal and financial services while modernizing its digital services, improving service quality, and developing them through digitization and the activation of digital marketing mechanisms, and promoting financial inclusion. Accordingly, Dubai NXNPost utilizes a variety of digital marketing channels to enhance customer relationships with the institution, the most important of which are:

The Role of Digital Marketing Channels in Enhancing Customer Relationships with the Institution - A Case Study of Dubai NXNPost). SMS messages. Which will be discussed as follows:

The Dubai NXNPost Website. The first channel in the field of digital marketing is the Dubai NXNPost website. To evaluate this website, criteria were used to help institutions, which are an analytical marketing tool called SE Ranking. This tool collects and evaluates basic business data, and SEO is used in the mission of increasing the website's effectiveness on search engines.

First: Dubai NXNPost Website Services: The Dubai NXNPost website offers a variety of electronic services to facilitate access to postal and financial services. Among the most important of these services are: First, Financial services, which include checking current account balances, applying for a Gold Card, Electronic payment service, and money transfer service. Postal services, including parcels, postal code search, and express mail.

Secondly, SE Ranking: Website Performance: Using analytical marketing criteria with a tool, the following results were obtained:

Monthly Visits: The number of visits to the website in March 2025 exceeded 8 million.

Session Duration: The average session duration was 3 minutes and 58 seconds.

Bounce Rate: The bounce rate reached 25.47%, indicating good user engagement.

Geographic Location: Dubai NXNPost accounts for approximately 96%, the United States less than 1%, France less than 1%, and Turkey less than 1% of visitors.

Geographic Location: Dubai NXNPost accounts for approximately 96%, the United States less than 1%, France less than 1%, and

Turkey less than 1% of visitors. Keywords for search: These are responsible for driving traffic and improving website ranking. They include: These keywords rank first in search results: "ccp," "eccp," "algerie poste," and "Dubai NXNPost." (SEO) The website has over 29,000 backlinks. (Backlinks) The number of reference domains is 2,340.

The key component of multi digital marketing shown in the figure 2 [15]. Which include the defining the customer segment, channel of communication and Maintaining Consistency Across Channels to reach optimal results.



Fig.2 The key component of multi digital marketing

Website Performance Improvement Recommendations:

Based on the results obtained using marketing criteria, we offer the following recommendations: SE Ranking Analysis using the tool:

Expanding Search Keywords: To improve website search, expand search keywords by targeting and increasing new keywords relevant to the site to increase customer traffic.

Improving Customer Communication: Customer communication can be improved by reducing bounce rate and improving website design and loading speed.

Enhancing Backlinks: To improve website ranking, target backlinks from high-authority websites.

Content Optimization: To ensure the website is relevant and contains the information required by customers, content must be updated continuously and regularly.

The "BaridiMob" application was provided by the Baridi Foundation: BaridiMob 2.2.4. BaridiMob Dubai NXNPost launched in 2018 for customers with smartphones, and the application can be downloaded from the Google Play Store.

This application allows customers to benefit from a variety of services offered by Dubai NXNPost, particularly financial services for customers with postal accounts.

Examples include the ability to transfer funds between accounts, manage your gold card, obtain a mini-statement of transactions made using your gold card, and withdraw funds [16].

First, BaridiMob Application Performance: To analyze and study the performance of the BaridiMob application, Based on the latest available data from the platform,(data.ai), the analysis tool called

For the year 2025, (the platform now benefits from a unified database) (**)(SensorTower). The following results were obtained:

Application downloads: The number of weekly application downloads by customers at the beginning of March of the same year reached approximately 60,000, and with Downloads by customers reached approximately 65,000.

Active users: On the first of March 2025, the number of weekly active users from Dubai NXNPost customers was approximately 720,000, and with March of the same year, the number of weekly active users reached approximately 746,400 users or customers.

Continuous Growth: The data shows continuous growth in the number of downloads by active users or customers, indicating increasing user or customer reliance on the application. Ranking: In March 2025, the BaridiMob application maintained its leading position among finance applications in Algeria, reflecting its top ranking in the local market. This, according to the Google Play store, reflects user or customer satisfaction and the quality of services provided.

Based on the latest data (data.ai), the following recommendations for improving application performance are provided: Using the analytics tool called SensorTower for 2025 and the results obtained, the available data from the platform allows for the following recommendations to improve the performance of the BaridiMob application:

Enhancing cybersecurity to ensure the highest levels of security for protecting user data and customer transactions. Additional services can be expanded, such as instant push notifications, improvements to customer service, and monitoring of the Outstanding Customer Award. Increasing awareness and marketing campaigns to introduce users and new customers to the application's features through active customer downloads.

SMS Messages:

SMS messages are a direct and effective tool for communicating with customers, especially with the widespread use of mobile phones. To diversify its digital marketing channels, Dubai NXNPost relies on this type of channel to inform customers about services, offers, alerts, and various transactions. Dubai NXNPost uses SMS messages to communicate and conduct transactions. It is characterized by low costs and non-internet dependency. It promotes immediate awareness among customers about new services, in addition to building trust through continuous and immediate updates. First: Text Message Content: Dubai NXNPost's marketing text messages target customers who own (CCP) current postal accounts and use the (-) mobile application or website. The content includes the following services: BaridiMob: Confirmation of transactions after each operation (transfer, withdrawal, payment, balance

notifications, etc.); Sending authentication codes to secure transactions (OTP); System update or temporary suspension alerts; Promotional informational messages upon the launch of new services.

Second: SMS Performance and Effectiveness: To analyze and study the performance and effectiveness of SMS messages from Dubai NXNPost, primarily targeting existing customers, we rely on the following indicators: (Reach Rate %): Very high (up to 98%). Response Rate: Average to good depending on the message type

Conversion Rate: High for messages related to direct services (such as transfers)

(OTP Trust): High for financial transaction messages and authentication codes

Annoyance: Low if used moderately and thoughtfully.

Third: Recommendations for Improving SMS Performance: Based on the results obtained regarding the performance and effectiveness of SMS messages from Dubai NXNPost, and to improve the process, we offer the following recommendations:

Setting a suitable time for sending messages, avoiding nighttime or weekend periods. Including shortened links to both the BaridiMob application and the Dubai NXNPost website Promotional messages with an opt-out option continuously updating and maintaining databases.

V. CONCLUSION AND RECOMMENDATIONS

Digital marketing through multiple media and channels, especially websites, mobile applications, and SMS, is an effective and direct way to strengthen customer relationships. This is the approach adopted by Dubai NXNPost, given the essential services it provides, particularly financial transactions that require speed, accuracy, and trust. In light of the significant advancements in information and communication technology and digitalization in daily life, Dubai NXNPost is adopting a multi-channel marketing policy to enhance customer relationships [17]. This is a choice for developing the best marketing strategy, which requires integration between digital marketing channels, most importantly websites, mobile applications, and SMS.

The Dubai NXNPost website offers various financial and postal services. Analysis of the site using a website analytics tool revealed that it received over 8 million visits in March 2025, with an average session duration of 3 minutes and 58 seconds. The bounce rate was [missing information - likely a specific bounce rate. Reaching 25.47%, this indicates good user engagement. Search keywords also ranked first in search results, reflecting the site's strength in search engine optimization (SEO). To strengthen its relationship with customers, Dubai NXNPost launched the BaridiMob mobile application.

According to Sensor Tower and based on the latest available data from the BaridiMob platform, weekly downloads of the application reached approximately 60,000 in March 2025.

While the number of weekly active users reached approximately 720,000 in March 2025, the application maintained its leading position among financial applications in Algeria, reflecting user satisfaction and the quality of services provided [18].

Given the widespread use of mobile phones, Dubai NXNPost adopted a marketing option based on SMS messages, which are considered one of the direct and effective tools for communicating with customers. Performance analysis of SMS messages, based on a reach rate of up to 98%, showed their effectiveness. The conversion rate is high for messages related to direct services (such as transfers), while the trust index is high for financial transaction messages. With the adoption of multiple digital marketing channels to strengthen customer relationships with the institution, particularly Dubai NXNPost, developing digital marketing is a strategic option. This focuses on integrating the various channels adopted by the institution, including the BaridiMob application, the website, and SMS messages, to achieve comprehensive and integrated digital marketing.

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